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4.1.1. Pandas - series creation and manipulation11.02

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

Test case 1

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:--

Odd:-- 5.0

Even:-- 6.0

dtype: float64

Test case 2

4 4 4 5 6 7 2 2 2 10 12 34 16

Mean of even and odd numbers:--

Even:-- 6.833333

dtype: float64

Test case 3

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 series = pd.Series(numbers)
8 # Grouping by even and odd numbers and calculating the mean
9 grouped = series.groupby(series % 2 == 0).mean()
10
11 # Display the mean of even and odd numbers with label
12 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
13 print("Mean of even and odd numbers:")
14 print(grouped)
15
```

Average time0.006 s

Maximum time0.008 s

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1

Expected output

Mean of even and odd numbers:--

Odd:-- 5.0

Even:-- 6.0

dtype: float64

Actual output

1 2 3 4 5 6 7 8 9 10

Mean of even and odd numbers:--

Odd:-- 5.0

Even:-- 6.0

dtype: float64

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4.1.2. Dictionary to dataframe10.22

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Sample Test Cases

datafram...

```
1 import pandas as pd
2
3 # Provided dictionary of list
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35],
7 }
8
9
10 # Convert the dictionary to a DataFrame
11 df = pd.DataFrame(data)
12
13 # Display the original DataFrame
14 print("Original DataFrame:")
15 print(df)
16
17 # Adding a new row
18 new_name = input("New name: ")
19 new_age = int(input("New age: "))
20 df.loc[len(df)] = [new_name, new_age]
21
22 # Display the DataFrame after adding a new row
```

Average time0.045 s

Maximum time0.049 s

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1

Expected output

Original DataFrame:

..... Name Age

0 Alice 25

1 Bob 30

2 Charlie 35

New name: Alice

Actual output

Original DataFrame:

..... Name Age

0 Alice 25

1 Bob 30

2 Charlie 35

New name: Alice

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4.1.3. Student Information

12.5s

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

Test case 1

studentinfo.txt

First five rows:

	Name	Age	Grade
0	John	25	A
1	Alin	22	B
2	Emu	24	A
3	Mike	23	C
4	Sony	21	B

Average age: 23.29

Students with a grade up to B:

	Name	Age	Grade
0	John	25	A
1	Alin	22	B
2	Emu	24	A

studentinfo.txt

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep="\\t", header=None, names=["Name", "Age", "Grade"])
6
7 # write your code here..
8 print("First five rows:")
9 print(data.head())
10 avg_age = round(data['Age'].mean(), 2)
11 print("Average age:", avg)
12 filtered_data = data[data['Grade'].isin(['A', 'B'])]
13 print("Students with a grade up to B")
14 print(filtered_data)
15
```

Average time0.017 s17.00 ms

Maximum time0.019 s19.00 ms

1 out of 1 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Test case 1

Expected output

studentinfo.txt

First five rows:

	Name	Age	Grade
0	John	25	A
1	Alin	22	B
2	Emu	24	A

Average age: 23.29

Students with a grade up to B:

	Name	Age	Grade
0	John	25	A
1	Alin	22	B
2	Emu	24	A

Actual output

studentinfo.txt

First five rows:

	Name	Age	Grade
0	John	25	A
1	Alin	22	B
2	Emu	24	A

TerminalTest Cases

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4.2.1. Month with the Highest Total Sales

03.1s

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Note:
The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

Test case 1

sales_data.csv

Best month: 2025-01

Total sales: \$1218.00

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 df['Sales'] = df['Quantity'] * df['Price']
9
10 df['Month'] = pd.to_datetime(df['Date']).dt.to_period('M')
11
12 monthly_sales = df.groupby('Month')['Sales'].sum()
13
14 # Find the month with the highest total sales
15 best_month = monthly_sales.idxmax()
16 highest_sales = monthly_sales.max()
17
18
19
20
21
```

Average time0.009 s9.33 ms

Maximum time0.011 s11.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

sales_data.csv

Best month: 2025-01

Total sales: \$1218.00

Actual output

sales_data.csv

Best month: 2025-01

Total sales: \$1218.00

TerminalTest Cases

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4.2.2. Best Selling Product

03:43

🔍📄🔗🔖

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2020-01-01	Product A	10	\$5	New York
2020-01-02	Product A	15	\$5	Los Angeles
2020-01-03	Product A	20	\$5	New York
2020-01-04	Product C	5	\$8	Chicago
2020-01-05	Product A	12	\$5	Chicago
2020-01-06	Product A	8	\$5	Los Angeles
2020-01-07	Product C	10	\$8	New York
2020-01-08	Product A	18	\$5	Los Angeles
2020-01-09	Product A	14	\$5	Chicago
2020-01-10	Product C	12	\$8	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

Test case 1

🔍📄🔗🔖

Expected output

🔍📄🔗🔖

Best selling product: Product A

Total quantity sold: 23

monthFor...

1

import pandas as pd

2

3

Prompt the user for the file name

4

file_name = input()

5

6

Load the data

7

df = pd.read_csv(file_name)

8

9

10

ps = df.groupby('Product')['Quantity'].sum()

11

Find the product with the highest total quantity sold

12

best_product = ps.idxmax()

13

highest_quantity = ps.max()

14

15

Display the result

16

print(f"Best selling product: {best_product}")

17

print(f"Total quantity sold: {highest_quantity}")

18

Average time

0.007 s

7.00 ms

Maximum time

0.008 s

8.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

🔍📄🔗🔖

Expected output

🔍📄🔗🔖

Best selling product: Product A

Total quantity sold: 23

Actual output

🔍📄🔗🔖

Best selling product: Product A

Total quantity sold: 23

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4.2.3. City that Sold the Most Products

03:15

🔍📄🔗🔖

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by city and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2020-01-01	Product A	10	\$5	New York
2020-01-02	Product A	15	\$5	Los Angeles
2020-01-03	Product A	20	\$5	New York
2020-01-04	Product C	5	\$8	Chicago
2020-01-05	Product A	12	\$5	Chicago
2020-01-06	Product A	8	\$5	Los Angeles
2020-01-07	Product C	10	\$8	New York
2020-01-08	Product A	18	\$5	Los Angeles
2020-01-09	Product A	14	\$5	Chicago
2020-01-10	Product C	12	\$8	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

Test case 1

🔍📄🔗🔖

Expected output

🔍📄🔗🔖

City sold the most products: Los Angeles

monthFor...

1

import pandas as pd

2

3

Prompt the user for the file name

4

file_name = input()

5

6

Load the data

7

df = pd.read_csv(file_name)

8

9

10

write the code..

11

citys = df.groupby('City')['Quantity'].sum()

12

best_city = citys.idxmax()

13

city = citys.max()

14

Display the result

15

print(f"City sold the most products: {best_city}")

16

Average time

0.008 s

8.33 ms

Maximum time

0.011 s

11.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

🔍📄🔗🔖

Expected output

🔍📄🔗🔖

City sold the most products: Los Angeles

Actual output

🔍📄🔗🔖

City sold the most products: Los Angeles

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4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date)

Output the product pairs that was sold most frequently

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,1,10,New York

2025-01-01,Product A,1,15,Los Angeles

2025-01-01,Product A,2,20,New York

2025-01-01,Product C,4,30,Chicago

2025-01-01,Product A,1,15,Chicago

2025-01-01,Product A,1,15,Los Angeles

2025-01-01,Product C,4,30,New York

2025-01-01,Product A,1,15,Los Angeles

2025-01-01,Product A,1,15,Chicago

2025-01-01,Product C,1,10,Los Angeles

Explanation:

Transactions:

2025-01-01: Product A, Product B

2025-01-02: Product A, Product C

2025-01-03: Product B, Product A

2025-01-04: Product C, Product B

2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

Product A and Product B: Appear in transactions on 2025-01-01 and 2025-01-03.

Product A and Product C: Appear in transactions on 2025-01-02 and 2025-01-05.

Product B and Product C: Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

Product A and Product B (2 times)

Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Explorer

1import pandas as pd

2from itertools import combinations

3from collections import Counter

4

5# Prompt user to input the file name

6file_name = input()

7

8# Read data from the specified CSV file

9df = pd.read_csv(file_name)

10

11# Write the code

12daily_transactions = df.groupby('Date')['Product'].apply(list)

13pair_counts = Counter()

14

15# 2. Iterate through each date's transaction

16for products in daily_transactions:

17 products = sorted(products)

18

19 # Generate all possible pairs (Combinations of size 2)

20 # If a date has fewer than 2 products, this produces nothing (correctly)

21 pairs = list(combinations(products, 2))

22

Average time0.008 s7.87 ms

Maximum time0.010 s10.00 ms

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

Product A and Product B: 2 times

Product A and Product B: 2 times

Product A and Product C: 2 times

Product A and Product C: 2 times

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4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.

2. Display the last 5 rows of the dataset.

3. Get the shape of the dataset (number of rows and columns).

4. Get a summary of the dataset (using .info()).

5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().

6. Check for missing values and display the count of missing values for each column.

7. Fill missing values in the 'Age' column with the median age.

8. Fill missing values in the 'Embarked' column with the most frequent value (mode()).

9. Drop the 'Cabin' column due to many missing values.

10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Harris	male	22	0	0	21517	7.25	5	
2	1	1	Mr. John Bradley	male	30	0	0	PC 17599	71.2833	C85	C
3	1	3	Mr. James Watson	male	36	0	0	3101	53.1	5	
4	1	3	Mr. James Watson	male	36	0	0	3101	53.1	5	
5	0	3	Mr. William Henry	male	36	0	0	3101	53.1	5	
6	0	3	Mr. William Henry	male	36	0	0	3101	53.1	5	
7	0	3	Mr. William Henry	male	36	0	0	3101	53.1	5	
8	0	3	Mr. William Henry	male	36	0	0	3101	53.1	5	
9	0	3	Mr. William Henry	male	36	0	0	3101	53.1	5	
10	1	1	Mr. John Bradley	male	30	0	0	PC 17599	71.2833	C85	C

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,Mr. Owen Harris,male,22,0,0,21517,7.25,5,

2,1,1,Mr. John Bradley,male,30,0,0,PC 17599,71.2833,C85,C

3,1,3,Mr. James Watson,male,36,0,0,3101,53.1,5,

4,1,3,Mr. James Watson,male,36,0,0,3101,53.1,5,

5,0,3,Mr. William Henry,male,36,0,0,3101,53.1,5,

6,0,3,Mr. William Henry,male,36,0,0,3101,53.1,5,

7,0,3,Mr. William Henry,male,36,0,0,3101,53.1,5,

8,0,3,Mr. William Henry,male,36,0,0,3101,53.1,5,

9,0,3,Mr. William Henry,male,36,0,0,3101,53.1,5,

10,1,1,Mr. John Bradley,male,30,0,0,PC 17599,71.2833,C85,C

Explorer

1import pandas as pd

2import numpy as np

3

4# Load the Titanic dataset

5data = pd.read_csv('Titanic-Dataset.csv')

6

7# 1. Display the first 5 rows of the dataset

8print(data.head())

9

10# 2. Display the last 5 rows of the dataset

11print(data.tail())

12

13# 3. Get the shape of the dataset

14print(data.shape)

15

16# 4. Get a summary of the dataset (info)

17print(data.info())

18

19# 5. Get basic statistics of the dataset

20print(data.describe())

21

Average time0.065 s65.00 ms

Maximum time0.065 s65.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

PassengerId Survived Pclass Fare Cabin Embarked

0 1 1 1 1 1 1 1 1 1 1 1

1 1 1 1 1 1 1 1 1 1 1 1

2 1 1 1 1 1 1 1 1 1 1 1

3 1 1 1 1 1 1 1 1 1 1 1

4 1 1 1 1 1 1 1 1 1 1 1

5 1 1 1 1 1 1 1 1 1 1 1

6 1 1 1 1 1 1 1 1 1 1 1

7 1 1 1 1 1 1 1 1 1 1 1

8 1 1 1 1 1 1 1 1 1 1 1

9 1 1 1 1 1 1 1 1 1 1 1

10 1 1 1 1 1 1 1 1 1 1 1

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4.2.6. Titanic Dataset Analysis and Data Cleaning -2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
- Convert the 'Sex' column to numeric values (male: 0, female: 1).
- One-hot encode the 'Embarked' column, dropping the first category.
- Get the mean age of passengers.
- Get the median fare of passengers.
- Get the number of passengers by class.
- Get the number of passengers by gender.
- Get the number of passengers by survival status.
- Calculate the survival rate of passengers.
- Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,Brown, Mr. Owen Harris, male, 30, 1, 0, A/5 21173, 7.25, 5
2,1,1, Gonnard, Mr. John Henry (Edward George Gonnard), female, 19, 1, 0, PC 17599, 71.00, C85, C
3,1,3, Hildesheim, Miss. Ida, female, 60, 0, 1, STON/O 18285, 53.00, 5
4,1,3, Colville, Mrs. Jacques Heath (Lily May Peel), female, 0, 1, 0, 16100, 53.1, C123, 5
5,0,3, Allen, Mr. William Henry, male, 35, 0, 0, 35968, 51.00, 5
6,0,3, Brown, Mr. James, male, 29, 0, 0, 5807, 51.00, Q
7,0,3, McCarthy, Mr. Timothy, male, 50, 0, 0, 1740, 53.00, 5
8,0,3, Talism, Master. Anni (Annie), male, 1, 1, 0, 9000, 51.00, 5
9,1,3, Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg), female, 37, 0, 1, 34742, 51.53, 5
10,1,3, Becker, Mrs. Nicholas (Adelie Helen), female, 19, 0, 0, 10770, 18.00, C
```

Note: Refer to the visible test case for better reference.

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SUBMIT

ticDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7
8 # 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise)
9 data['IsAlone'] = (data['FamilySize'] == 0).astype(int)
10
11 # 2. Convert 'Sex' to numeric (male: 0, female: 1)
12 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
13
14 # 3. One-hot encode the 'Embarked' column
15 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
16
17 # 4. Get the mean age of passengers
18 print(data['Age'].mean())
19
20
21 # 5. Get the median fare of passengers
```

Average time: 0.018 s, Maximum time: 0.018 s, 1 out of 1 shown test case(s) passed

Test case 1

Expected output: 29.69911764705882, 14.4542, 3... 401, 1... 216, 2... 104

Actual output: 29.69911764705882, 14.4542, 3... 401, 1... 216, 2... 104

Name: Pclass, dtype: int64, Name: Pclass, dtype: int64

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4.2.7. Titanic Dataset Analysis and Data Cleaning -3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarkation location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,Brown, Mr. Owen Harris, male, 30, 1, 0, A/5 21173, 7.25, 5
2,1,1, Gonnard, Mr. John Henry (Edward George Gonnard), female, 19, 1, 0, PC 17599, 71.00, C85, C
3,1,3, Hildesheim, Miss. Ida, female, 60, 0, 1, STON/O 18285, 53.00, 5
4,1,3, Colville, Mrs. Jacques Heath (Lily May Peel), female, 0, 1, 0, 16100, 53.1, C123, 5
5,0,3, Allen, Mr. William Henry, male, 35, 0, 0, 35968, 51.00, 5
6,0,3, Brown, Mr. James, male, 29, 0, 0, 5807, 51.00, Q
7,0,3, McCarthy, Mr. Timothy, male, 50, 0, 0, 1740, 53.00, 5
8,0,3, Talism, Master. Anni (Annie), male, 1, 1, 0, 9000, 51.00, 5
9,1,3, Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg), female, 37, 0, 1, 34742, 51.53, 5
10,1,3, Becker, Mrs. Nicholas (Adelie Helen), female, 19, 0, 0, 10770, 18.00, C
```

Note: Refer to the visible test case for better reference.

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ticDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10 # 1. Calculate the survival rate by class
11 print(data.groupby('Pclass')['Survived'].mean())
12 print(data.groupby('Embarked_S')['Survived'].mean())
13 print(data.groupby('FamilySize')['Survived'].mean())
14
15 # 4. Calculate the survival rate by being alone
16 print(data.groupby('IsAlone')['Survived'].mean())
17
18 # 5. Get the average fare by class
19 print(data.groupby('Pclass')['Fare'].mean())
20
21 # 6. Get the average age by class
```

Average time: 0.024 s, Maximum time: 0.024 s, 1 out of 1 shown test case(s) passed

Test case 1

Expected output: Pclass: 1... 0.629630, 2... 0.472826, 3... 0.242363

Actual output: Pclass: 1... 0.629630, 2... 0.472826, 3... 0.242363

Name: Survived, dtype: float64, Name: Survived, dtype: float64

Embarked_S: 1... 0.629630, 2... 0.472826, 3... 0.242363

Terminal, Test Cases

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

0122

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Get the number of survivors by gender (Sex).
- Get the number of non-survivors by gender (Sex).
- Get the number of survivors by embarkation location (Embarked_S).
- Get the number of non-survivors by embarkation location (Embarked_S).
- Calculate the percentage of children (Age < 18) who survived.
- Calculate the percentage of adults (Age >= 18) who survived.
- Get the median age of survivors.
- Get the median age of non-survivors.
- Get the median fare of survivors.
- Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	PClass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, PClass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21173, 7.25, 5
2, 1, 1, "Cumings, Mrs. John Bradley (Lucy Deane) Briggs", female, 38, 1, 0, PC 17599, 71.00, C85, C
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O 3101282, 53.1, 5
4, 1, 1, "Nye, Mr. Jacobus Mathias (Lily May Peck)", female, 32, 1, 0, 3101282, 53.1, C123, 5
5, 0, 3, "Allen, Mr. William Henry", male, 35, 0, 0, 3101282, 53.1, 5
6, 0, 3, "Brown, Mr. James", male, 35, 0, 0, 3101282, 53.1, 5
7, 0, 3, "Carter, Mr. Thomas", male, 35, 0, 0, 3101282, 53.1, 5
8, 0, 3, "Falcon, Master. Anna (young)", male, 35, 0, 0, 3101282, 53.1, 5
9, 0, 3, "Henderson, Mrs. Oscar W. (Elizabeth Vilhelmina Berg)", female, 37, 0, 0, 3101282, 53.1, 5
10, 0, 3, "Hewson, Mrs. William (Ada Helen)", female, 35, 0, 0, 3101282, 53.1, 5
```

Note: Refer to the visible test case for better reference.

titanicDat...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8
9
10 # 1. Get the number of survivors by gender
11 survivors = data[data['Survived'] == 1]
12 print(survivors['Sex'].value_counts())
13
14 # 2. Get the number of non-survivors by gender
15 non_survivors = data[data['Survived'] == 0]
16 print(non_survivors['Sex'].value_counts())
17
18 # 3. Get the number of survivors by embarked location
19 print(survivors['Embarked_S'].value_counts())
20
21 # 4. Get the number of non-survivors by embarked location
22 print(non_survivors['Embarked_S'].value_counts())
23
24 # 5. Calculate the percentage of children (Age < 18) who survived
25 children = data[data['Age'] < 18]
26 child_survival_rate = children['Survived'].mean()
```

Average time: 0.018 s

Maximum time: 0.018 s

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

female ... 233

male ... 109

female ... 233

male ... 109

Name: Sex, dtype: int64

female ... 408

male ... 408

Name: Sex, dtype: int64

female ... 81

male ... 81

Name: Sex, dtype: int64

Terminal

Test Cases

PREV

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NEXT